

The Establishment of a Portfolio of Companies

---Simulation of Modeling and Optimising

UCD MIS3003S BBS41FT Business Analytics Project

Introduction

The U.S. stock market experienced a period of significant volatility following the introduction of sweeping trade tariffs. By the end of May 2025, the stock market has returned to pre-crash levels. Given that, to profit from a strong market recovery, a portfolio of companies is constructed to stimulate price behavior and optimize a two-week investment strategy, from Aug 04 to Aug 15, 2025.

The portfolio is composed of eight companies listed on the NASDAQ Stock Exchange across different sectors for the purpose of risk diversification, including Zenas BioPharma, Inc. (NASDAQ: ZBIO), Zillow Group, Inc. (NASDAQ: Z), Zebra Technologies Corporation (NASDAQ: ZBRA), Zions Bancorporation, National Association (NASDAQ: ZION), HIVE Digital Technologies Ltd. (NASDAQ: HIVE), Hut 8 Corp. (NASDAQ: HUT), Cadence Design Systems, Inc. (NASDAQ: CDNS), and CSX Corporation (NASDAQ: CSX).

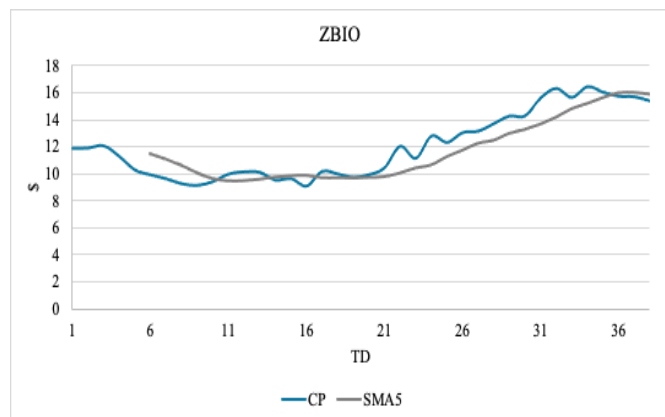
The analysis is structured into three sequential phases: descriptive, predictive, and prescriptive analytics. For each company, historical price showing a positive slope for the observed period, from Jun 09 to Aug 01, 2025, is discussed. After that, simple regression models to predict the closing price (CP) are performed, combined with critical metrics. These models reveal strong upward trends for most stocks, with Zebra and Cadence exhibiting the highest daily price (DG). High R^2 values of Zebra (0.8814) and CXS (0.8869) indicate a strong model fit, while the relatively high RMSE value of Cadence (11.2707) indicates a significant degree of volatility of the stock.

The prescriptive part focuses on formulating and solving an integer linear programming (ILP) optimisation model to maximise the portfolio's total expected return based on certain constraints. The optimal allocation invested nearly the entire budget (\$159,998.1800) and is proposed to create a return of \$6,962.8220. The solution allocated significantly capital to HIVE, Zenas, and Hut due to higher expected return per dollar. Further, the observed profit was \$7,582.1100 during the investment period, outperforming the model's prediction by 8.89%, largely due to outperformance by Zebra and Hut.

Zenas BioPharma, Inc. (NASDAQ: ZBIO)

As a clinical stage biopharmaceutical corporation, Zenas is committed to developing and commercializing transformative therapies in immunology and inflammation for patients living with autoimmune diseases. The company's lead product candidate, obexelimab, is the first bifunctional B cell targeted antibody for the treatment of IgG4-related diseases (IgG4-RD) (Zenas, 2025).

Although ZBIO's CP and SMA5 of exhibited certain volatility throughout the observed period, there was a pure long-term upward trend according to its time-series plot. This appreciation can be attributed to sustained market demand for novel autoimmune treatment and positive clinical perceptions of obexelimab's differentiated mechanism, such as well-tolerated profile (Zenas, 2025).



Specifically, the CP showed a slightly declined trend from \$11.92 to \$9.19 per share during the first then trading days, due to negative news surrounding a Class Action Lawsuit and the business's high cash burn. Subsequently, the stock price increased significantly, surging 64.18% to a peak of \$16.45 by the 34th trading day, primarily driven by optimistic investor sentiment related to the upcoming completion of the Phase 3 INDIGO trial and the Phase 2 SunStone trial, which was indicated significant upside potential for the stock.

A simple linear regression model was applied to predict the CP, resulting in the equation:

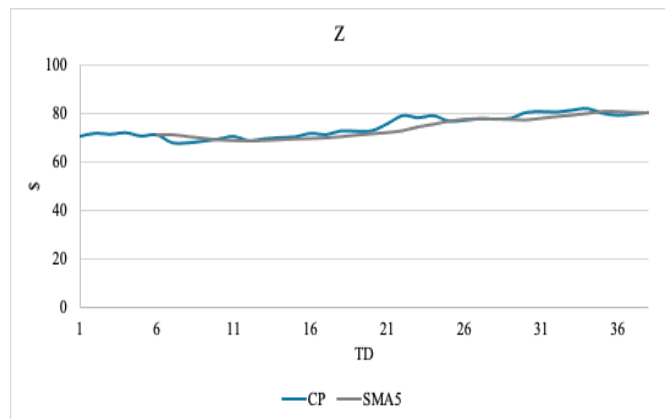
$$CP = 8.6539 + 0.1749 * TD$$

- Daily price gain (DG): The slope coefficient of 0.1749 means that the CP would raise by an average of \$0.1749 for each additional TD.
- RMSE: A value of 1.5003 is considered to have relatively significant volatility despite of a clear upward trend.
- R^2 : A value of 0.6331 means that around 63.31% of the variation in Zenas's CP can be explained by the linear passage of TD.
- BSR: A value of 0.6667 (#22/33) indicates that more than half of CP data remained above the SMA5, which is typically considered a bullish signal.

Zillow Group, Inc. (NASDAQ: Z)

Zillow is an Internet-based real estate firm that generates revenue by providing seamless solutions across residential purchases, mortgages, and rentals. The company is committed not only to enhancing the home shopping experience via tech tools such as 3D Home tours, but also to capturing buyer attention with Zillow Showcase. Its flagship online marketplace, Zillow.com, features comprehensive for-sale and rental listings, while its subsidiary, Zillow Home Loans, offers mortgage pre-approvals and financing services (Zillow, 2025).

The CP exhibited an overall a consistent upward trend over the entire plotted period. Beginning at \$70.28, it gradually increased and stabilized at a higher level as trading days progressed, despite minor fluctuations. This can be attributed to Zillow's portfolio of large and trusted brands (e.g., Trulia and StreetEasy portals), its living database of homes, as well as relatively strong financial position (Zillow, 2025).



Further, the short-term average price movement aligned closely with the daily CP trend, demonstrating no considerable daily variations. The minor fluctuations in CP (e.g., TD 6-11) were not enough to significantly lower the SMA5, which means that the upward momentum is relatively sustained.

A simple linear regression model was applied to predict the CP, which as follows:

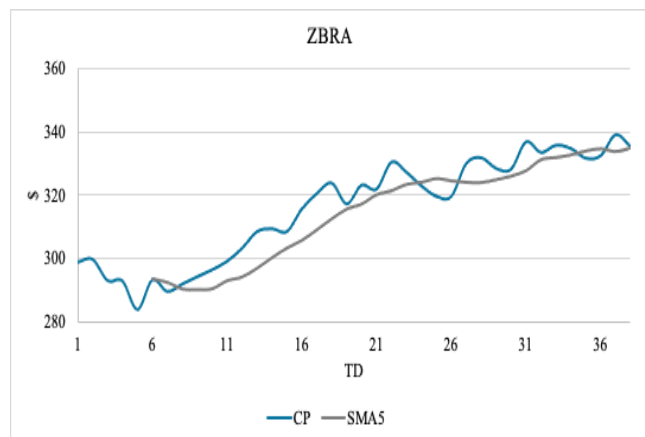
$$CP = 67.1023 + 0.3734 * TD$$

- Daily price gain (DG): The slope coefficient of 0.3734 indicates that the CP would raise by an average of \$0.3734 for each additional TD.
- RMSE: A value of 2.1129 is considered to have significant volatility around the trend line, indicating typical the U.S. residential real estate market variability due to inflationary conditions and housing availability .
- R²: A value of 0.7986 means that around 79.86% of the variation in Zillow's CP can be explained by the linear passage of TD, reflecting a good fit of the model.
- BSR: A value of 0.6061 (#20/33) indicates that the CP remained above the SMA5 for more than half of the period. Combined with the entire upward trajectory, there is a bullish short-term sentiment for the stock.

Zebra Technologies Corporation (NASDAQ: ZBRA)

Zebra is a leading provider of automatic identification and data capture technology, serving more than 80% of the Fortune 500 businesses. The company operates in over 100 countries, with primary focus on the retail, transportation logistics, healthcare, and manufacturing markets. And its product portfolio consists of barcode scanners and printers, workflow optimization software, and mobile computing devices. Benefiting from scale advantages, Zebra maintains a highly diversified business across business segments, end markets, and geographies (Zebra, 2025).

The CP and SMA5 of Zebra displayed a clear upward trend over the plotted period, despite an initial slight decrease due to market fluctuations. As of the 38th TD, the CP had increased by 12.26% to \$335.24. This appreciation was driven by strong business performance and a positive full-year sales growth outlook (Simply Wall St, 2025), including increased earnings reported for the second quarter of 2025. Further, for most



of trading days, the daily CP traded above the SMA5, which is a potential positive signal in terms of upward trend of its stock price, despite short-term adjustment attributed to intense market competition and evolving customer demands.

A simple linear regression model was applied to predict the CP, as shown below:

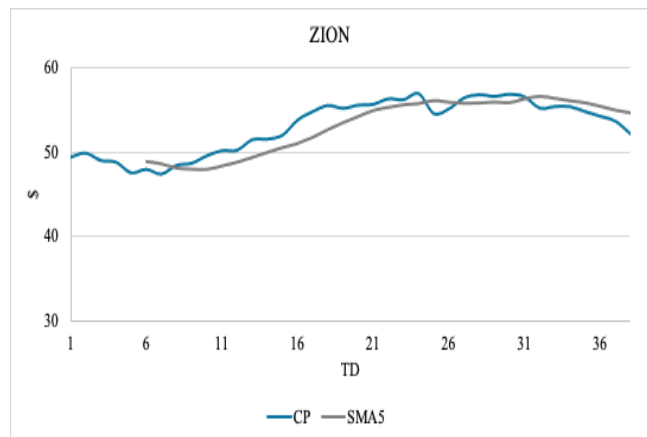
$$CP = 288.3619 + 1.4031 * TD$$

- Daily price gain (DG): The relatively high slope coefficient of 1.4031 indicates that the CP would raise by an average of \$1.4031 for each additional TD.
- RMSE: A value of 5.7987 suggests that the daily stock price is quite volatile, despite of a clear upward trend.
- R^2 : A value of 0.8814 means that around 88.14% of the variation in Zebra's CP can be explained by the linear passage of TD, meaning a good fit of the model.
- BSR: A value of 0.7879 (#26/33) confirms that the CP remained above the SMA5 for the majority of the period, reinforcing the presence of a short-term upward trend.

Zions Bancorporation, National Association (NASDAQ: ZION)

Headquartered in Salt Lake City, Utah, Zions provides a wide range of banking and financial services. Its offerings include commercial and small business banking, capital markets and investment banking, commercial real estate lending, retail banking, and wealth management. As of 2024, the company operated 404 branches with 9,406 full-time employees, mainly across the state of California, Arizona, and New Mexico. Further, Zions generates revenue through its seven distinct bank divisions, which are organized by geographic area (Zions, 2025).

The CP and SMA5 of Zions demonstrated a clear upward trend over the entire plotted period. In the initial stage (TD 1-7), the CP of the stock experienced a slight decrease, due to uncertain macroeconomic outlook on the company's commercial loan portfolio. Subsequently, the CP surged by 20.24%, from \$47.28 (TD 7) to \$56.85 (TD 8), which was primarily driven by 2Q25 earnings beat expectations. For instance, Zions' net earnings for the second quarter of 2025 reached \$243 million, compared to \$190 million in 2Q24, with diluted EPS of \$1.63 (2Q24: \$1.28) (Zions, 2025). However, a slight downward trend emerged in the third stage (TD 32-38), which was attributed to persistent increases in operating expenses and the competitive banking environment.



A simple linear regression model was applied to predict the CP, as defined by the equation:

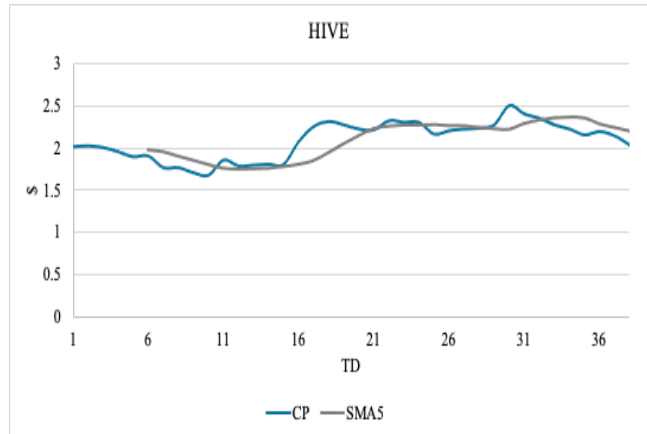
$$CP = 48.5426 + 0.2285 * TD$$

- Daily price gain (DG): The positive slope coefficient of 0.2285 indicates that the CP would increase by an average of \$0.2285 for each additional TD.
- RMSE: A value of 2.0190 signifies considerable volatility around the trend line. And this consistent with the economic uncertainty affecting banks that specialize in commercial lending.
- R^2 : A value of 0.6191 means that about 61.91% of the variation in Zions's CP can be explained by the linear passage of TD, suggesting a moderately acceptable fit of the model.
- BSR: A value of 0.6667 (#22/33) means that more than half of the CP was above the SMA5, signaling a prevailing short-term bullish sentiment for the stock.

HIVE Digital Technologies Ltd. (NASDAQ: HIVE)

HIVE is a technology company focused on constructing and operating data centers driven by green and clean energy in Bermuda. The company not only engages in mining and sale digital currencies and performance computing hosting, but also provides infrastructure solutions for blockchain and AI applications based on its cutting-edge data centers in Paraguay, Canada, and Sweden (HIVE , 2025).

HIVE's stock price demonstrated an upward trend, starting around \$2.01 (TD 1) and ending around \$2.03 (TD 38), while there was intraday fluctuations during the observed period. A primary driver of positive investor sentiment was the anticipated completion of the acquisition of the Yguazú site , which was expected to raise HIVE's mining capability and hashrate (HIVE, 2025). However, the stock's volatility also reveal broader market challenges, including volatile Bitcoin prices and increasing costs of mining, which directly affect the company's revenue and stock performance.



A simple linear regression model was applied to predict the CP, defined by the following equation:

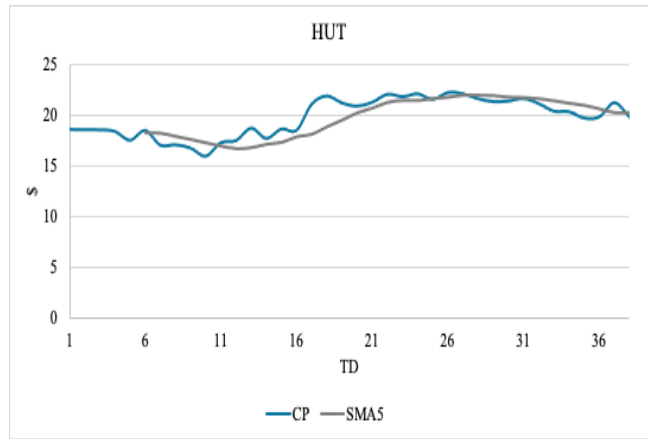
$$CP = 1.8231 + 0.0135 * TD$$

- Daily price gain (DG): The positive slope coefficient of 0.0135 indicates that the CP would rise by an average of \$0.0135 for each additional TD, although the trend is modest.
- RMSE: A value of 0.1675 is quite low, meaning a relatively small average variance between the model's predicted values and the actual values.
- R²: A value of 0.4509 means that about 45.09% of the variation in HIVE's CP can be explained by the linear passage of TD.
- BSR: A value of 0.5152 (#17/33) means that more than half of the CP was above the SMA5, indicating a potential short-term bullish sentiment.

Hut 8 Corp. (NASDAQ: HUT)

Founded in 2017, Hut is a data mining firm engaged in energy infrastructure and Bitcoin in North American. The company generates revenue by operating in three main segments: Power (e.g., Power Generation), Digital Infrastructure (e.g., ASIC and CPU Colocation), and Compute (e.g., Bitcoin Mining). Hut also provides customers with supporting services, including software automation, process design, hosting operations, as well as finance and accounting (Hut, 2025).

As shown in the time-series plot, the CP experienced a decline with mild fluctuations during the initial phase (TD 1-10), but generally remained around a range of \$15-\$19. Afterwards, the CP began a sustained upward movement (TD 11-26), climbing from \$15.89 to a range between \$17 and \$22, and the SMA5 closely tracked the daily CP. This appreciation coincided with the



company's strong revenue report. In the second quarter of 2025, Hut realized a revenue of \$41.29 million, a year-on-year growth of 17.28%, which was driven by the ASIC fleet upgrades before launching American Bitcoin, a major-owned subsidiary of Hut (Hut, 2025). Following the peak, the CP entered a mild downward trend, retreating from \$22.24 to around \$20, aligning with a broader cooling trend in the Bitcoin market (e.g., BTC-USD).

To predict the CP, a simple linear regression model was applied, which as follows:

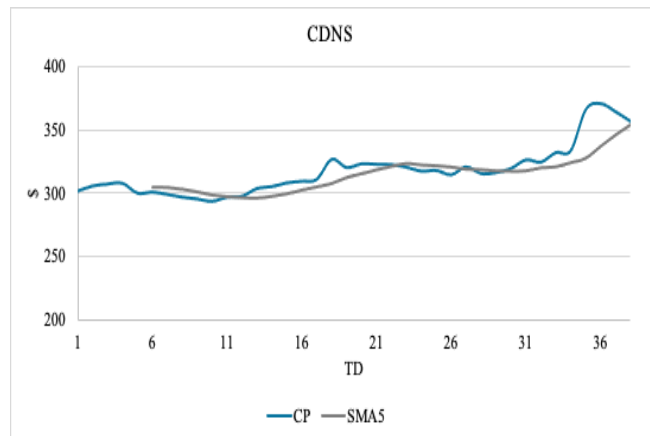
$$CP = 17.5031 + 0.1163 * TD$$

- Daily price gain (DG): The positive slope coefficient of 0.1163 indicates that the CP would rise by an average of \$0.1163 for each additional TD, despite that the tendency is relatively modest.
- RMSE: A value of 1.3982 means the model's prediction has a moderate deviation from actual prices, with a period of volatility.
- R^2 : A value of 0.4675 means that about 46.75% of the variation in Hut's CP can be explained by the linear passage of TD, and the model goodness-of-fit is not extremely impressive.
- BSR: A value of 0.5455 (#18/33) means that more than half of the CP was above the SMA5, with a potential short-term bullish sentiment.

Cadence Design Systems, Inc. (NASDAQ: CDNS)

Cadence is an industry-leading electronic system design company. Its products include electronic design automation (EDA) software, digital integrated circuit (IC) design and analysis, and intellectual property. The company also offers services in relation to methodology, education, and hosted design solutions. Leveraging its Intelligent System Design™ strategy, Cadence's design solutions serves a wide range of market, comprising aerospace, mobile communications, automotive, hyperscale computing, and life sciences (Cadence, 2025).

The CP and SMA5 of the stock exhibited a sustained upward trajectory throughout the entire plotted period. The CP remained relatively stable with minor fluctuations during the initial stage (TD 1-12). This was followed by a period of gradual and moderate momentum (TD 13-34). After that, the growth accelerated considerably, reaching \$366.26 by the end of period, which was attributed to robust



financial results. In the second quarter of 2025, Cadence reported a revenue of \$1.28 billion, an increase by 20% year-on-year, exceeding analysts' expectations. Further, some financial institutions raised their target price for the stock, signifying market optimism in terms of the company's continued performance (Bohen, 2025).

A simple linear regression model was applied to predict the CP, which as follows:

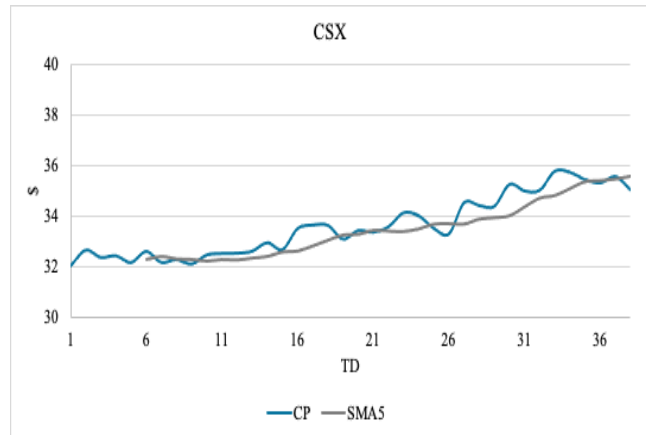
$$CP = 289.4251 + 1.4528 * TD$$

- Daily price gain (DG): The positive slope coefficient of 1.4528 indicates that the CP would rise by an average of \$1.4528 for each additional TD.
- RMSE: A value of 11.2707 means a significant degree of volatility, as the model's prediction has a substantial deviation from actual prices.
- R^2 : A value of 0.6784 indicates that 67.84% of the difference in Cadence's CP can be explained by the linear variation of TD. And the goodness-of-fit of the regression model is acceptable.
- BSR: A value of 0.6364 (#21/33) reveals that the CP has been higher than the SMA5 for closely two-thirds of the time, that is, a prolonged short-term bullish sentiment.

CSX Corporation (NASDAQ: CSX)

Founded in 1980, CSX provides rail-based freight transportation services across North American. Its extensive network serves nearly two-thirds of the U.S. populations. CSX generates revenue by operating four primary segments: merchandise, intermodal, coal, and trucking. The merchandise, which is the largest revenue contributor (FY24: \$8.90 billion), focuses on shipments of agriculture and food products, chemicals, minerals, and metals and equipment. The intermodal business is committed to providing a cost-effective and environmentally advantageous offerings to long-haul trucking (CSX, 2025).

The CP and SMA5 demonstrated a clear upward trend over the entire plotted period, with an intraday gain of 9.48%, despite certain fluctuations. Over the past five fiscal years, CSX's earnings per share grew at a compound annual growth rate (CAGR) of 4.4%, which is positive indicator for its stock valuation. Further, Barclays raised CSX's target price by 3.60% from \$31 to \$36, maintaining an overweight rating. The adjustment reveals a positive market outlook for the company's future performance (Tracker, 2025).



A simple linear regression model was applied to predict the CP, defined by the following equation:

$$CP = 31.5940 + 0.1016 * TD$$

- Daily price gain (DG): The positive slope coefficient of 0.1016 indicates that the CP would rise by an average of \$0.1016 for each additional TD, meaning a modest but positive trend.
- RMSE: A value of 0.4090 is relatively low, meaning a small average variance between the model's predicted values and the actual observed values.
- R^2 : A value of 0.8869 means that about 88.69% of the variation in CSX's CP can be explained by the linear passage of TD, with a strong goodness-of-fit.
- BSR: A value of 0.7273 (#24/33) means there is a sustained potential for short-term bullish signal.

Prescriptive Analytics: Optimisation

1. Optimisation Setup

1.1 Objective Function

The objective of optimisation is to maximise the total expected return of the investment over the required holding period, and the function is formulated as:

$$\max \sum DG_i * Shares_i * days_i$$

where DG_i is the daily price gain of a share of company i , $Shares_i$ is the number of shares to purchase in company i , and $days_i$ is the number of days invested for company i (i.e., $days_i = BSR_i * 10$).

1.2 Decision Variables

For each company i , $i \in \{ZBIO, Z, ZBRA, ZION, HIVE, HUT, CDNS, CSX\}$, the variable $Shares_i$ refers to the number of shares to buy.

1.3 Constraints

Three constraints are built to control the risk of the investment strategy, which as follows:

i. Per-company investment cap: The investment in any company i should be no more than $USD\ 50,000 * R^2$ based on the CP data for TD=38. The constraint is:

$$CP_i * Shares_i \leq USD\ 50,000 * R_i^2$$

ii. Budget cap: The total investment across the portfolio of 8 firms should not exceed $USD\ 20,000 * 8$. The constraint is:

$$\sum CP_i * Shares_i \leq \$160,000$$

iii. Integrality and non-negativity of $Shares_i$: The number of shares for each company should be a whole number, and the data should be more than zero. The constraint is:

$$Shares_i \geq 0 \cap Shares_i = integer$$

2. The Solution Proposed

The model is resolved using an integer linear programming (ILP) in Excel Solver, i.e., Simplex LP as a solving method. It yields a feasible allocation that maximises the total expected return based on the defined constraints.

Ticker	CP	R ²	DG	BSR	Days_Invested	Per-Co Cap (\$)	Shares (#)	Invested (\$)	Expected Return (\$)	Port. Cap (\$)
ZBIO	15.41	0.6331	0.1749	0.6667	7	31,656.7258	2,054	31,652.1400	2,515.2926	160,000.0000
Z	80.20	0.7986	0.3734	0.6061	7	39,929.5850	480	38,496.0000	1,254.7260	
ZBRA	335.24	0.8814	1.4031	0.7879	8	44,070.0979	131	43,916.4400	1,470.4659	
ZION	52.08	0.6191	0.2285	0.6667	7	30,956.7653	1	52.0800	1.5994	
HIVE	2.03	0.4509	0.0135	0.5152	6	22,546.9388	11,104	22,541.1200	897.4083	
HUT	19.78	0.4675	0.1163	0.5455	6	23,376.0427	1,180	23,340.4000	823.3299	
CDNS	356.97	0.6784	1.4528	0.6364	7	33,917.8476	0	0.0000	0.0000	
CSX	35.01	0.8869	0.1016	0.7273	8	44,342.7092	0	0.0000	0.0000	
								159,998.1800	6,962.8220	

The total portfolio investment sums to \$159,998.18, which is under the \$160,000 budget across all firms. The total expected return is computed to be \$6,962.82. Further, investments in ZBIO, Z, ZBRA, HIVE, and HUT approach their respective per-company investment ceilings. The model allocates no capital to CDNS and CSX, as their objective value per dollar (i.e., $DG_i * days_i / CP_i$) are quite low given the limited budget. In addition, ZION is allocated only one share due to relatively low expected return per dollar.

3. Predictive and Observed Profit

3.1 Deviation between predictive and observed profit

The optimization model predicts a total profit of \$6,962.82, which is the linear objective constructed from model coefficient DG_i and holding days $days_i$. During the investment period (Aug 04, 2025 to 15 Aug, 2025), the observed profit was \$7,582.11, which is calculated by summing the difference between the close price on the last day and the open price on the 4 Aug, 2025, multiplied by the number of shares bought for per company.

Around 8.89% higher observed profit over the holding period than the predictive profit indicates the model is conservative to predict the CP for the period. And the deviation is primarily driven by outperformance in ZBRA and HUT.

Ticker	Expected Return (\$)	Actual Return (\$)	Deviation (\$)
ZBIO	2,515.2926	2,464.8000	-50.4926
Z	1,254.7260	1,891.2000	636.4740
ZBRA	1,470.4659	-2,614.7600	-4,085.2259
ZION	1.5994	0.1900	-1.4094
HIVE	897.4083	1,887.6800	990.2717
HUT	823.3299	3,953.0000	3,129.6701
CDNS	0.0000	0.0000	0.0000
CSX	0.0000	0.0000	0.0000
	6,962.8220	7,582.1100	619.2880
			8.89%

3.2 Limitations

Linearity of the objective: The model assumes expected returns increase linearly and scales proportionally with shares based on the method of linear programming, which is a simplification that does not account for the market of large orders and non-linear price changes. Further, DG is considered as a constant over the observed period. If DG is estimated from the nearest window, then regime shifts (e.g., news and macro economy) may break the mapping from projected returns to actual returns.

Overfitting: Linking the per-company budget to R^2 will result in overfitting in terms of in-sample data, as a high R^2 does not guarantee the performance outside the sample. And any slight change in the input data will cause drastically change in the model.

Solution constraints: The integer linear programming results a set of discrete points, and this non-convexity reveals that local optimal solution may not necessarily be the global optimal solution, which creates a certain degree of error.

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